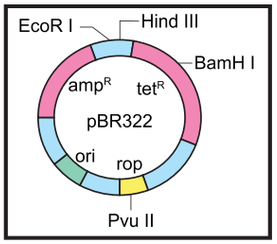
14. In genetic engineering, gene of interest is transferred to the host cell through a vector. Consider the following four agents (1-4) in this regard and select the correct option about which one or more of these can be used as vectors
1. A bacterium 2. Plasmid
3. Plasmodium 4. Bacteriophage
(AIPMT Main 2010)
a) 1 and 4 only b) **2 and 4 only**
c) 1 only d) 1 and 3 only

15. Given below is a sample of a portion of DNA strand giving the base sequence on the opposite strands. What is so special shown in it? (AIPMT 2014)
5'---GAATTC---3' 3'---CTTAAG---5'
a) **Palindromic sequence of base pairs**
b) Replication completed
c) Deletion mutation
d) Start codon at the 5'end

16. There is a restriction endonuclease called EcoRI. What does "co" part in it stand for ? (AIPMT 2011)
a) Coelom b) Colon
c) **Coli** d) Coenzyme

17. The figure below is the diagrammatic representation of the vector pBR322. Which one of the given options correctly identifies its certain components? (AIPMT 2012)



a) Ori-original restriction enzyme
b) rop-reduced osmotic pressure
c) Hind III, EcoRI – selectable markers
d) **ampR, tetR – antibiotic resistance genes**

18. A mixture containing DNA fragments a,b,c,d with molecular weights of a+b=c, a>b and d>c, was subjected to agarose gel electrophoresis. The position of these fragmets from cathode to anode sides of the gel would be (DPMT 2010)
a) **b,a,c,d** b) a,b,c,d
c) c,b,a,d d) b,a,d,c

19. An analysis of chromosomal DNA using the southern hybridisation technique does not use (AIPMT 2014)
a) Electrophoresis
b) Blotting
c) Autoradiography
d) **PCR**

20. The colonies of recombinant bacteria appear white in contrast to blue colonies of non- recombinant bacteria because of (NEET 2013)
a) Non-recombinant bacteria containing beta galactosidase
b) Insertional inactivation of a-galactosidase in non-recombinant bacteria
c) **Insertional inactivation of b-galactosidase in recombinant bacteria**
d) Inactivation of glycosidase enzyme in recombinant bacteria

21. Which one of the following palindromic base sequence in DNA can be easily cut at about the middle by some particular restriction enzyme? (AIPMT 2010)
a) 5'CGTTCG3' 3'ATCGTA 5'
b) 5' GATATG 3' 3' CTACTA 5'
c) **5' GAATTC 3' 3' CTTAAG 5'**
d) 5' CACGTA 3' 3' CTCAGT 5'

22. Silencing of mRNA has been used in producing transgenic plants resistant to (AIPMT, 2011)
a) Boll worms b) **Nematodes**
c) White rusts d) Bacterial blights

23. Some of the characteristics of Bt cotton are (AIPMT,2010)
a) Long fibre and resistant to aphids

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